AURA Project Build Log

Track A The Escalation Referee | Bản chốt 14/10/2026 | Phạm Gia Phú

AURA is a supervised reimbursement-review assistant. A vision model extracts structured receipt facts, deterministic C# policy decides the route, and people remain responsible for exceptions. The system covers upload, durable background processing, review, Audit, and Undo.

How I used AI

Qwen3.8 Flash through OpenRouter and Alibaba, with reasoning disabled, reads one JPG or PNG and returns a strict JSON contract. Codex helped trace code paths, design tests, review refactors, and synchronize documentation. I accepted changes only after build, tests, and evidence review. The model never receives approval authority.

Where AI helped

- Fast code navigation exposed the difference between an in-memory signal and the durable SQL job queue.
- Generated test cases expanded coverage for field roles, arithmetic, date evidence, fallback, concurrency, Audit, and UI contracts.
- Vision extraction handled varied receipt layouts while policy remained replaceable and testable.

Where AI cost time or money

Provider names and model slugs changed, and earlier routes produced 404 or intermittent 429 errors. Schema-valid outputs also contained semantic errors such as the wrong money unit or identifier role. A reasoning-enabled request used its completion budget without returning usable JSON. These failures required provider diagnostics, a strict schema, semantic validation, a single repair attempt, cooldown, and explicit claim boundaries. Paid API reruns were avoided once data became known.

What changed after failure

The first real-receipt holdout exposed a missed escalation even though the response parsed. I replaced “valid JSON is enough” with evidence contract v2: identifiers have distinct roles, dates preserve raw evidence, discounts are explicit, and an amount can auto-approve only when a printed final-total line corroborates it. Unresolved uncertainty becomes FACT review. The policy priority remains `FACT -> POLICY -> AUTHORITY`.

Verified evidence

Gate	Current result
Release build	0 warnings, 0 errors
Offline tests	163/163 passed
Live Verify	5/5 in 65.004 seconds
Prospective mini-holdout	8/10, 0 missed, 2 over, 0 system/provider/429 errors
Sequential known-data repeat	11/12, 0 missed, 1 over; stability evidence only

The prospective sample is small and redacted. It supports a supervised demo, not production accuracy. Three real-user sessions remain incomplete.

Largest feature cut

I cut production evidence localization from the final baseline. Highlighting where a fact appears could improve reviewer confidence, but without word-level OCR, independent annotations, and a false-highlight benchmark it could point to the wrong amount and create false confidence. A synthetic hash-bound SVG prototype remains isolated from extraction and policy. PDF, production RBAC, malware scanning, object storage, and distributed rate limiting also remain roadmap items.

Next gate

Complete three consented user sessions, preserve verbatim feedback, trace one change to a commit, and report one negative operational impact. Then reduce over-escalation with independent positive and negative fixtures without weakening the missed-escalation guard.